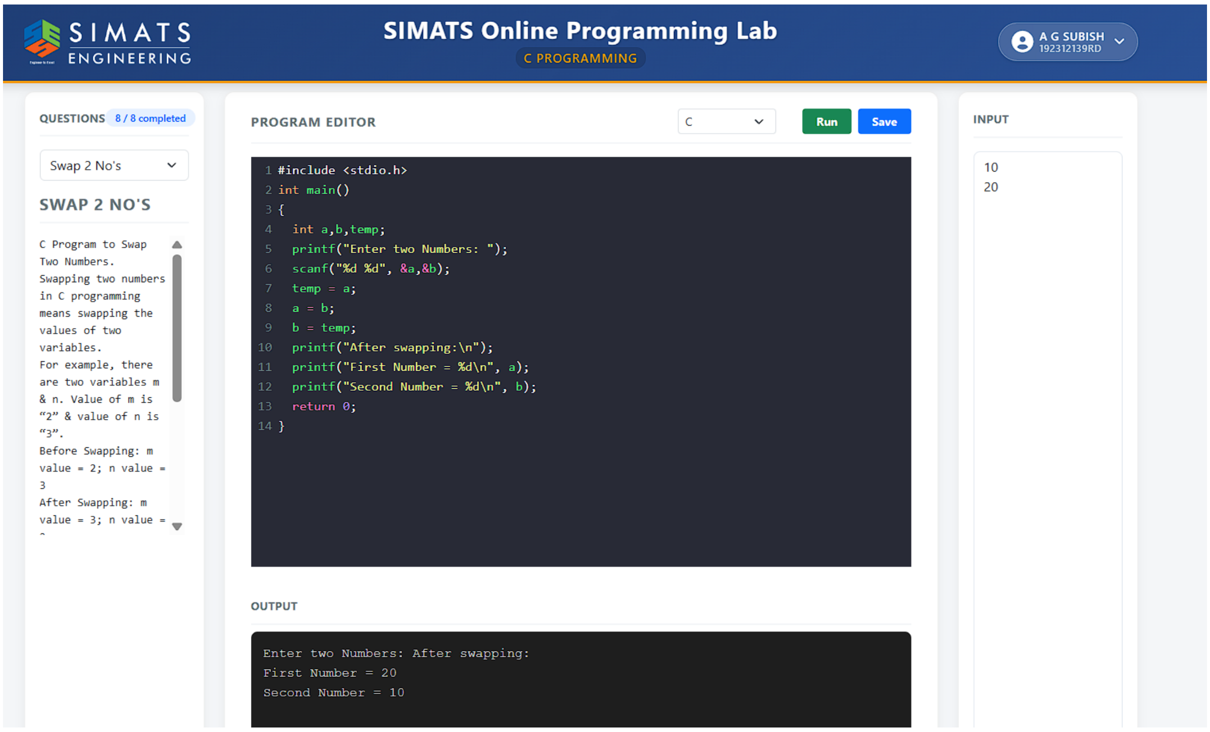
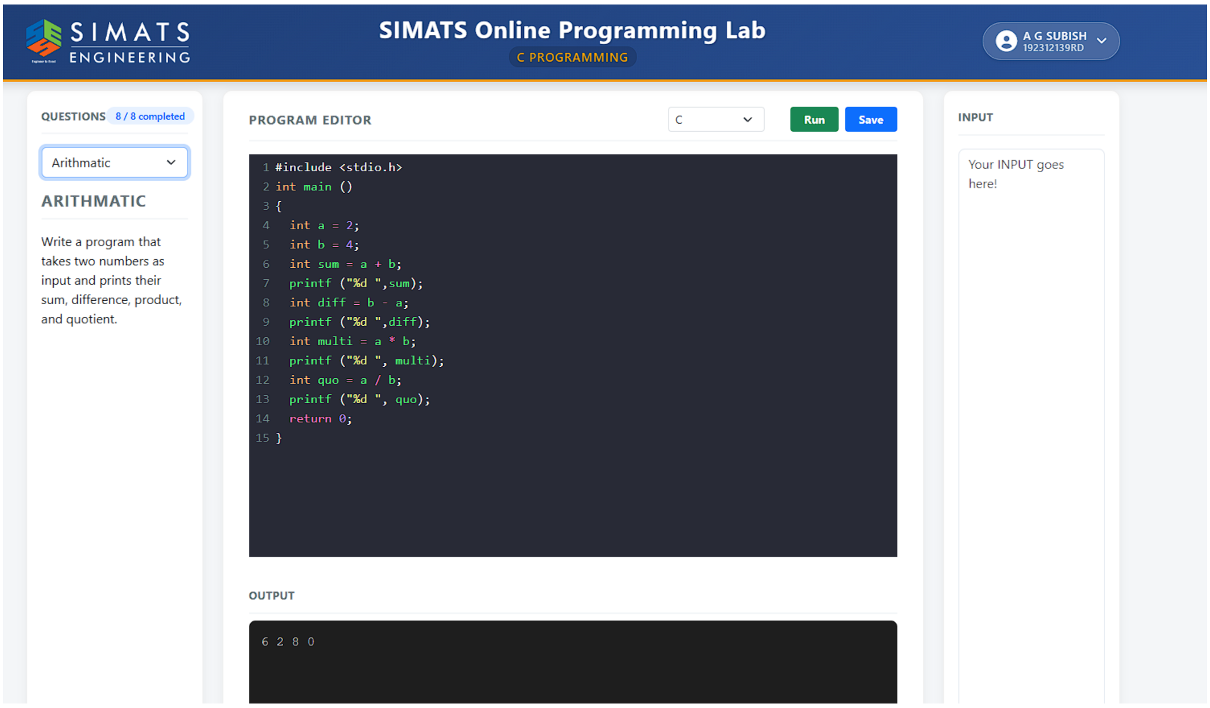
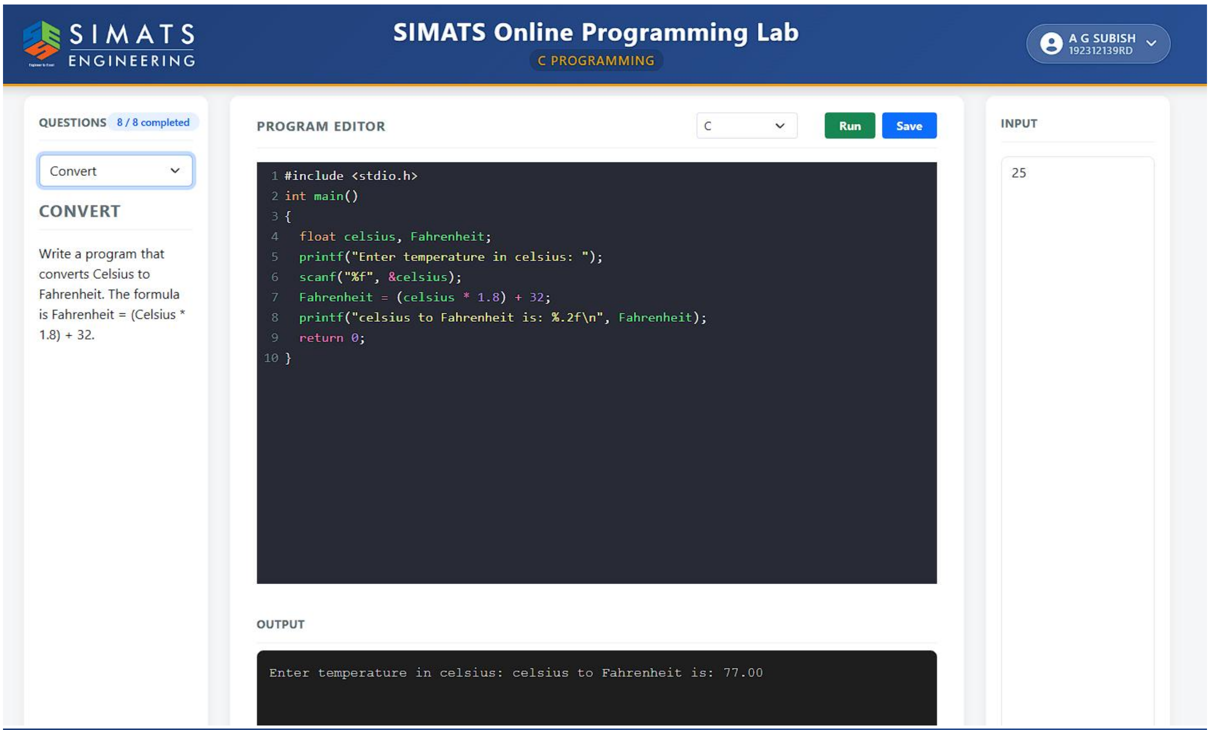
Operators and Expressions



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QUESTIONS
8 / 8 completed
Convert
CONVERT
Write a program that converts Celsius to Fahrenheit. The formula is Fahrenheit = (Celsius * 1.8) + 32.
PROGRAM EDITOR
C
Run
Save
#include <stdio.h>
int main()
{
float celsius, Fahrenheit;
printf("Enter temperature in celsius: ");
scanf("%f", &celsius);
Fahrenheit = (celsius * 1.8) + 32;
printf("celsius to Fahrenheit is: %.2f\n", Fahrenheit);
return 0;
}
OUTPUT
Enter temperature in celsius: celsius to Fahrenheit is: 77.00
INPUT
25


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QUESTIONS
8 / 8 completed
Rectangle
RECTANGLE
Write a program that calculates the area of a rectangle given its length and width.
Problem Description: Write a program that calculates the area of a rectangle given its length and width. The program should prompt the user to enter the length and width of the rectangle as input, then calculate the area
PROGRAM EDITOR
C
Run
Save
#include <stdio.h>
int main()
{
float celsius, Fahrenheit;
printf("Enter temperature in celsius: ");
scanf("%f", &celsius);
Fahrenheit = (celsius * 1.8) + 32;
printf("celsius to Fahrenheit is: %.2f\n", Fahrenheit);
return 0;
}
OUTPUT
Enter temperature in celsius: celsius to Fahrenheit is: 77.00
INPUT
25

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QUESTIONS 8 / 8 completed

ODD EVEN

ODD EVEN

Write a program that checks whether a number is even or odd.

PROGRAM EDITOR

C

Run Save

```c
#include <stdio.h>
int main()
{
  int num;
  printf("Enter the value: \n");
  scanf("%d", &num);
  if (num % 2 == 0)
    printf("%d is Even\n", num);
  else
    printf("%d is Odd\n", num);
  return 0;
}
```

INPUT

10

OUTPUT

```
Enter the value:
10 is Even
```

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QUESTIONS 8 / 8 completed

Leap

LEAP

Write a program that determines whether a year is a leap year or not. A year is a leap year if it is divisible by 4, but not divisible by 100 unless it is also divisible by 400.

PROGRAM EDITOR

C

Run Save

```c
#include <stdio.h>
int main()
{
  int year;
  printf("Enter a Year:\n");
  scanf("%d", &year);
  if ((year % 400 == 0) || (year % 4 == 0 && year % 100 != 0))
    printf("%d is a Leap Year:\n", year);
  else
    printf("%d is not a Leap Year:\n", year);
  return 0;
}
```

INPUT

2028

OUTPUT

```
Enter a Year:
2028 is a Leap Year:
```

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QUESTIONS 8 / 8 completed

Max-Min

MAX-MIN

Write a program that takes three integers as input and prints the maximum and minimum of the three.

PROGRAM EDITOR

C

Run Save

```c
#include <stdio.h>
int main ()
{
  int a,b,c,max,min;
  printf("Enter three integer value: \n");
  scanf("%d %d %d", &a, &b, &c);

  max = a;
  if (b > max)
      max = b;
  if (c > max)
     max = c;

  min = a;
  if (b < min)
    min = b;
  if (c < min)
    min = c;
  printf("Maximun value: %d\n", max);
  printf("Minimum value: %d\n", min);
  return 0;
```

OUTPUT

```
Enter three integer value:
Maximun value: 30
Minimum value: 10
```

INPUT

```
10
25
30
```

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QUESTIONS 8 / 8 completed

Sum-Avg

SUM-AVG

Write a program that takes a list of numbers as input and computes the sum and average of the numbers.

PROGRAM EDITOR

C

Run Save

```c
#include <stdio.h>
int main()
{
  int n,i;
  float sum = 0,avg;
  printf("Enter the number of elements: \n");
  scanf("%d", &n);
  float num[n];
  printf("Enter numbers %d: \n", n);
    for (i = 0; i < n; i++){
      scanf("%f", &num[i]);
      sum += num[i];
    }
  avg = sum / n;
  printf("sum = %.2f\n", sum);
  printf("Average = %.2f\n", avg);
  return 0;
}
```

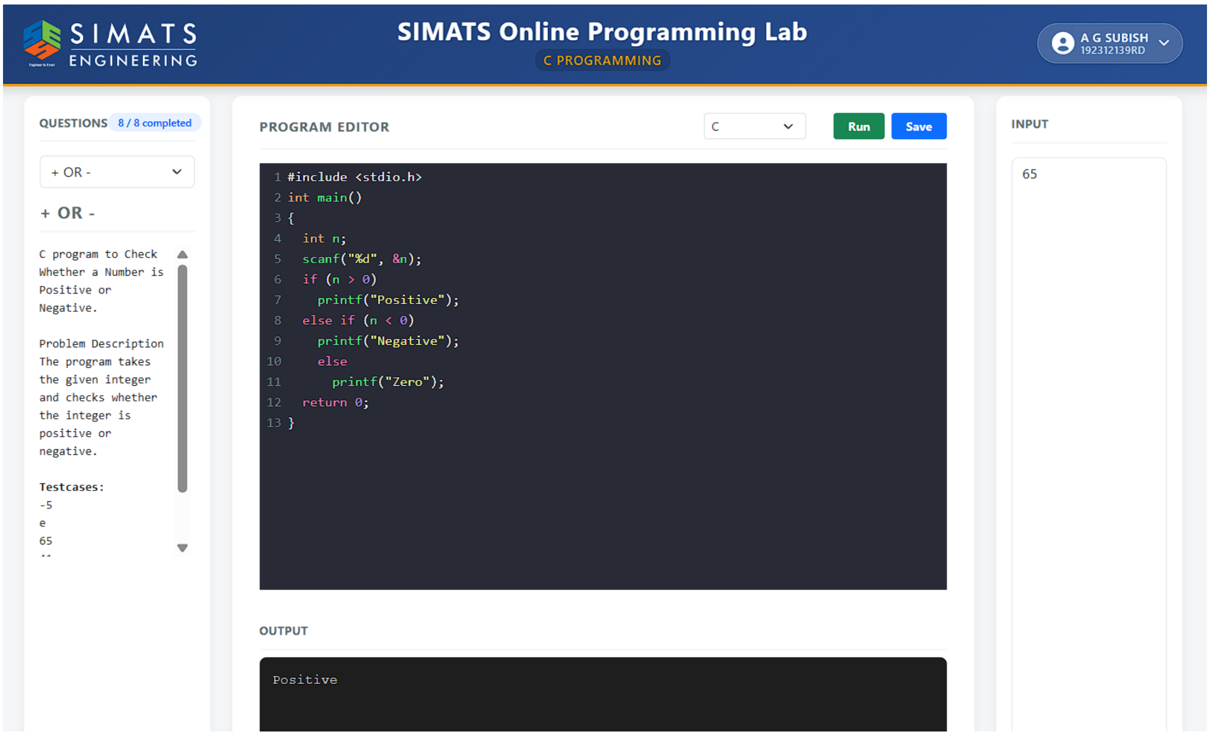
OUTPUT

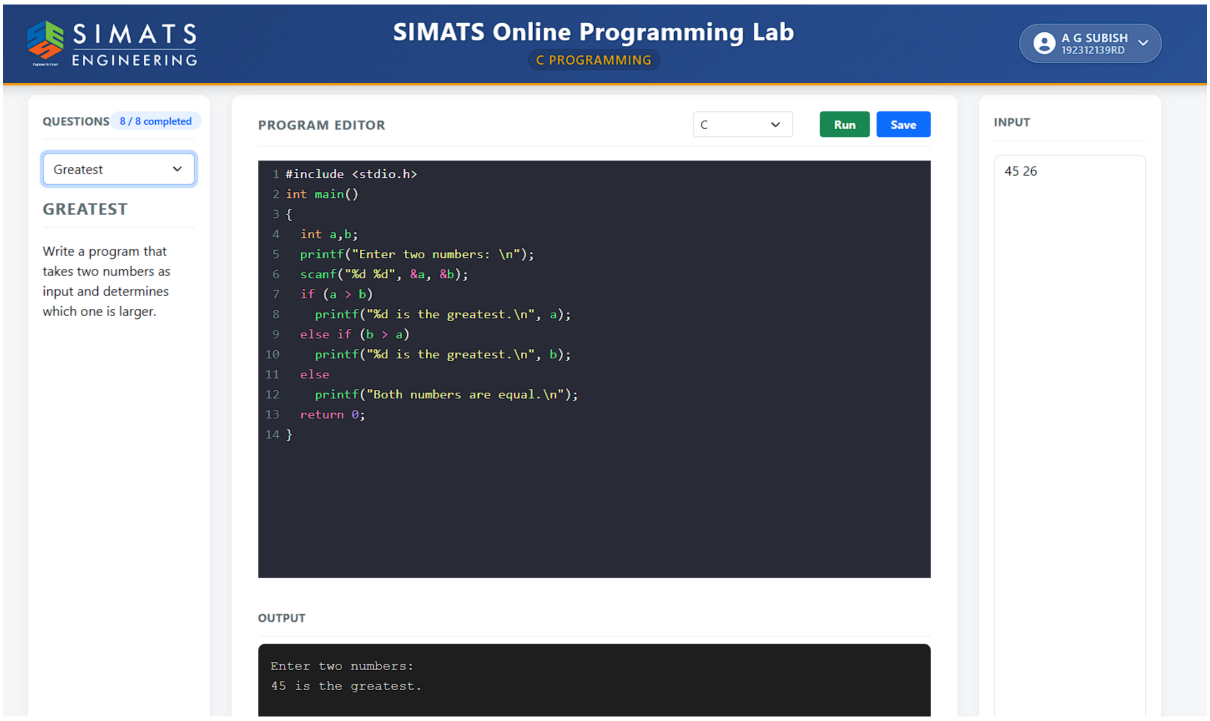
```
Enter the number of elements:
Enter numbers 5:
sum = 150.00
Average = 30.00
```

INPUT

```
5
10 20 30 40 50
```

Decision Macking





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QUESTIONS 8 / 8 completed

Character

CHARACTER

Write a program that takes a character as input and determines whether it is a vowel or a consonant.

PROGRAM EDITOR

C

Run Save

```c
#include <stdio.h>
int main()
{
  char ch;
  printf("Enter a character: \n");
  scanf("%c", &ch);
  if (ch== 'a' || ch== 'e' || ch== 'i' || ch== 'o' || ch== 'u' || ch== 'A' ||
      ch== 'E' || ch== 'I' || ch== 'O' || ch== 'U')
  {
    printf("%c is a Vowel\n", ch);
  }
  else
  {
    printf("%c is a consonant\n", ch);
  }
  return 0;
}
```

OUTPUT

```
Enter a character:
A is a Vowel
```

INPUT

A

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QUESTIONS 8 / 8 completed

Age

AGE

Write a program that takes an age as input and determines whether the person is a child, teenager, adult, or senior citizen.

PROGRAM EDITOR

C

Run Save

```c
#include <stdio.h>
int main()
{
  int age;
  printf("Enter age: \n");
  scanf("%d", &age);
  if (age < 13)
    printf("Child\n");
  else if (age >= 13 && age <= 19)
    printf("Teenager\n");
  else if (age >= 20 && age <= 59)
    printf("Adult\n");
  else
    printf("Senior Citizen\n");
  return 0;
}
```

OUTPUT

```
Enter age:
Adult
```

INPUT

25

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QUESTIONS 8 / 8 completed

Temperature

TEMPERATURE

Write a program that takes a temperature as input and determines whether it is hot, warm, or cold.

PROGRAM EDITOR

C

Run Save

```c
#include <stdio.h>
int main()
{
  float temp;
  printf("Enter Temperature (c): \n");
  scanf("%f", &temp);
  if (temp >= 30)
    printf("Hot\n");
  else if (temp >= 20)
    printf("Warm\n");
  else
    printf("Cold\n");
  return 0;
}
```

OUTPUT

```
Enter Temperature (c):
Warm
```

INPUT

34

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QUESTIONS 8 / 8 completed

Ups-Lows

UPS-LOWS

Write a program that takes a character as input and determines whether it is uppercase or lowercase.

PROGRAM EDITOR

C

Run Save

```c
#include <stdio.h>
int main()
{
  char ch;
  printf("Enter a character: \n");
  scanf("%c", &ch);
  if (ch >= 'A' && ch <= 'Z')
    printf("%c is an uppercase letter.\n", ch);
  else if (ch >= 'a' && ch <= 'z')
    printf("%c is a lowercase letter.\n", ch);
  else
    printf("%c is not an alphabet.\n", ch);
  return 0;
}
```

OUTPUT

```
Enter a character: G is an uppercase letter.
```

INPUT

G

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QUESTIONS 8 / 8 completed

Multiple

MULTIPLE

Write a program that takes a number as input and determines whether it is a multiple of 3 or 5.

PROGRAM EDITOR

C Run Save

```c
#include <stdio.h>
int main()
{
  int num;
  printf("Enter a number: \n");
  scanf("%d", &num);
  if (num % 3 == 0 && num % 5 == 0)
    printf("%d is a multiple of both 3 and 5.\n", num);
  else if (num % 3 == 0)
    printf("%d is a multiple of 3.\n", num);
  else if (num % 5 == 0)
    printf("%d is a multiple of 5.\n", num);
  else
    printf("%d is not a multiple of 3 or 5.\n", num);
  return 0;
}
```

OUTPUT

```
Enter a number: 15 is a multiple of both 3 and 5.
```

INPUT

15

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QUESTIONS 8 / 8 completed

Grade

GRADE

Write a program that takes a grade as input and determines the corresponding letter grade (A, B, C, D, or F).

PROGRAM EDITOR

C Run Save

```c
#include <stdio.h>
int main()
{
  int grade;
  printf("Enter the marks (0-100): \n");
  scanf("%d", &grade);
  if (grade >= 90 && grade <= 100)
    printf("Grade = A\n");
  else if (grade >= 80)
    printf("Grade = B\n");
  else if (grade >= 70)
    printf("Grade = C\n");
  else if (grade >= 60)
    printf("Grade = D\n");
  else if (grade >= 0)
    printf("Grade = F\n");
  else
    printf("Invalid Marks\n");
  return 0;
}
```

OUTPUT

```
Enter the marks (0-100):
Grade = C
```

INPUT

75

Looping


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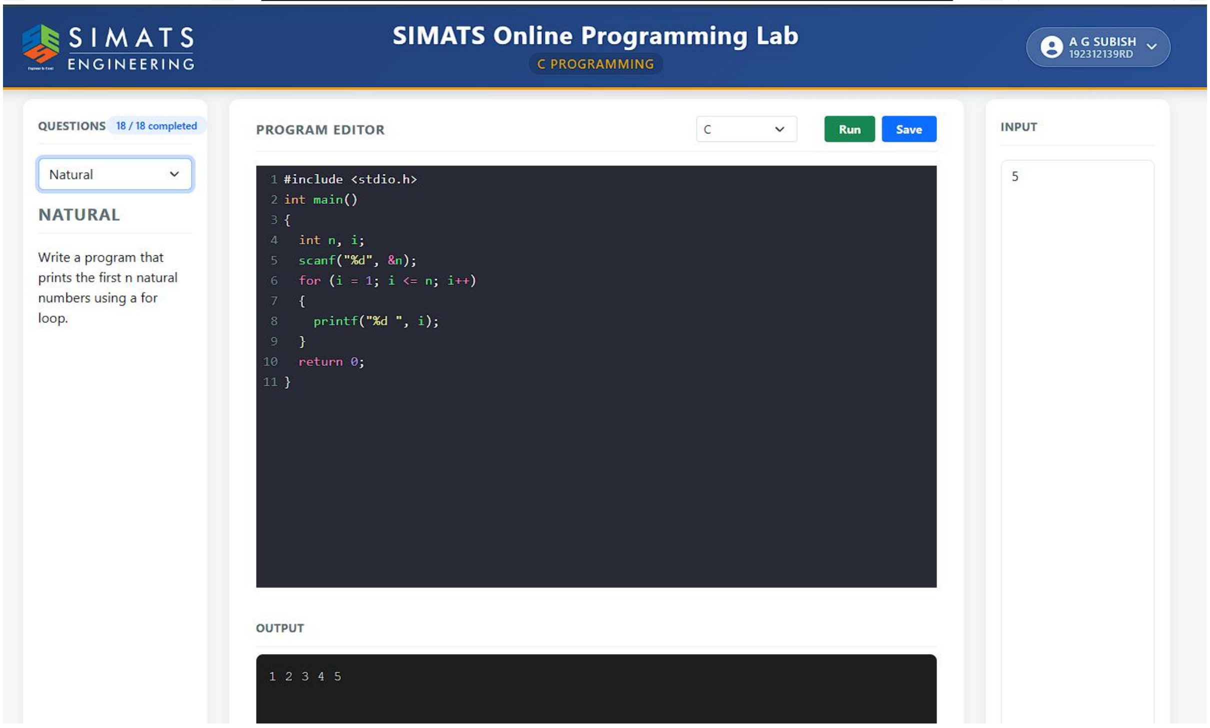
QUESTIONS
18 / 18 completed
Sum(ODD,EVEN)
SUM(ODD,EVEN)
C Program to Find the Sum of Even and Odd Numbers.
Problem Description
The program takes the number N and finds the sum of odd and even numbers from 1 to N.
Testcases:
-5
10
4
100
PROGRAM EDITOR
C
Run
Save
#include <stdio.h>
int main()
{
int N, i;
int oddSum = 0, evenSum = 0;
printf("Enter the value of N: \n");
scanf("%d", &N);
for (i = 1; i <= N; i++)
{
if (i % 2 == 0)
evenSum = evenSum + i;
else
oddSum = oddSum + i;
}
printf("Sum of Even Numbers = %d\n", evenSum);
printf("Sum of Odd Numbers = %d\n", oddSum);
return 0;
}
OUTPUT
Enter the value of N:
Sum of Even Numbers = 650
Sum of Odd Numbers = 625
INPUT
50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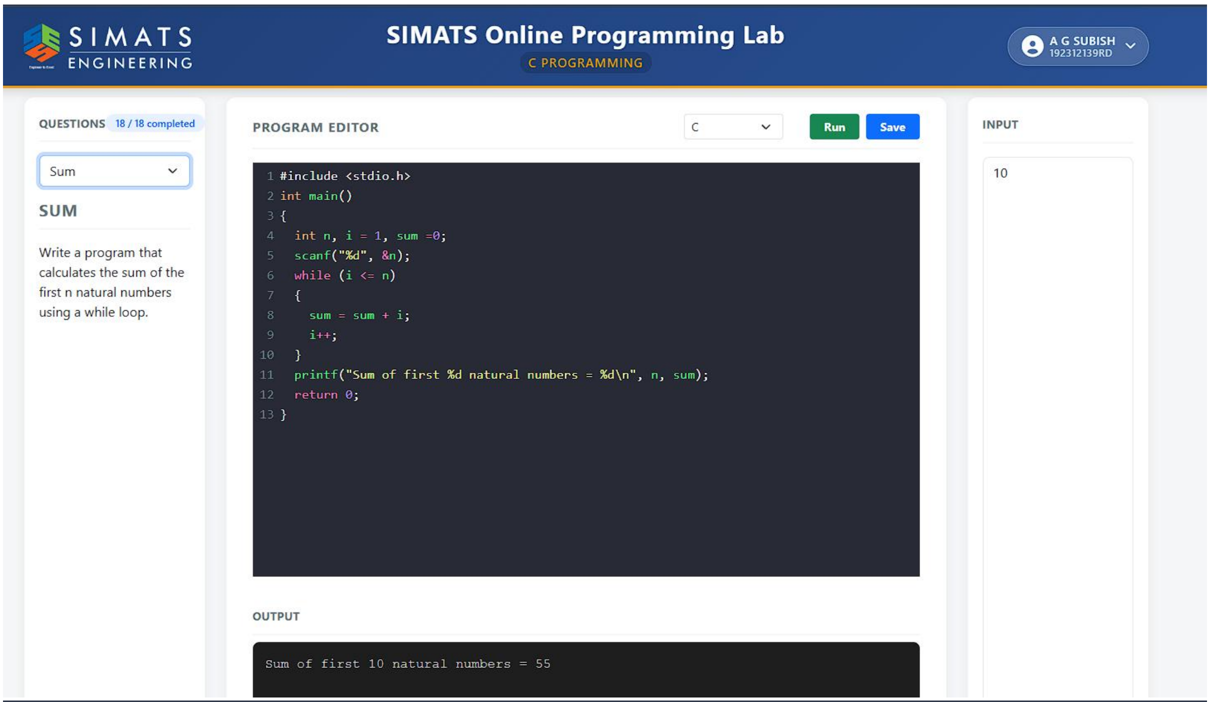
QUESTIONS
18 / 18 completed
Natural
NATURAL
Write a program that prints the first n natural numbers using a for loop.
PROGRAM EDITOR
C
Run
Save
#include <stdio.h>
int main()
{
int n, i;
scanf("%d", &n);
for (i = 1; i <= n; i++)
{
printf("%d ", i);
}
return 0;
}
OUTPUT
1 2 3 4 5
INPUT
5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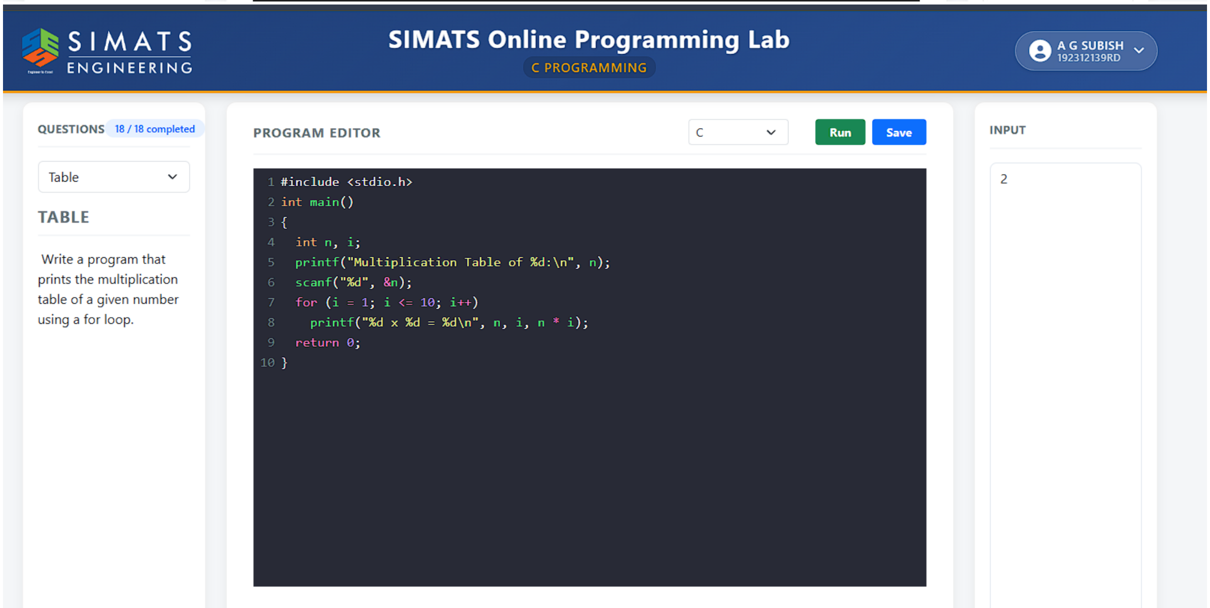
QUESTIONS 18 / 18 completed
Sum
SUM
Write a program that calculates the sum of the first n natural numbers using a while loop.
PROGRAM EDITOR
C
Run
Save
#include <stdio.h>
int main()
{
int n, i = 1, sum =0;
scanf("%d", &n);
while (i <= n)
{
sum = sum + i;
i++;
}
printf("Sum of first %d natural numbers = %d\n", n, sum);
return 0;
}
OUTPUT
Sum of first 10 natural numbers = 55
INPUT
10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QUESTIONS 18 / 18 completed
Table
TABLE
Write a program that prints the multiplication table of a given number using a for loop.
PROGRAM EDITOR
C
Run
Save
#include <stdio.h>
int main()
{
int n, i;
printf("Multiplication Table of %d:\n", n);
scanf("%d", &n);
for (i = 1; i <= 10; i++)
printf("%d x %d = %d\n", n, i, n * i);
return 0;
}
INPUT
2

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QUESTIONS 18 / 18 completed

Prime

PRIME

Write a program that takes a number as input and determines whether it is a prime number or not using a while loop.

PROGRAM EDITOR

C

Run Save

```c
#include <stdio.h>
int main()
{
  int n, i = 2, prime = 1;
  printf("Enter a number: \n");
  scanf("%d", &n);
  if (n <= 1)
  {
    prime = 0;
  }
  else
  {
    while (i <= n / 2)
    {
      if (n % i == 0)
      {
        prime = 0;
      }
      i++;
    }
  }
```

OUTPUT

```
Enter a number:
13 is a prime Number.
```

INPUT

13

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QUESTIONS 18 / 18 completed

Count Vows

COUNT VOWS

Write a program that takes a string as input and counts the number of vowels in the string using a for loop.

PROGRAM EDITOR

C

Run Save

```c
#include <stdio.h>
int main()
{
  char str [100];
  char ch;
  int i, count = 0;
  printf("Enter a String: \n");
  scanf("%[^\n]", str);
  for (i = 0; str[i] != 0; i++)
  {
    ch = str[i];
    if (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' ||
        ch == 'u' || ch == 'A' || ch == 'E' || ch == 'I' ||
        ch == 'O' || ch == 'U')
             count++;
    }
  printf("Number of vowels = %d", count);
  return 0;
}
```

OUTPUT

```
Enter a String:
Number of vowels = 2
```

INPUT

Hello

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QUESTIONS 18 / 18 completed

Fibonacci

FIBONACCI

Write a program that prints the Fibonacci series up to a given number using a while loop.

PROGRAM EDITOR C Run Save

```c
#include <stdio.h>
int main()
{
  int n, i, first = 0, second = 1, third = 0;
  scanf("%d", &n);
  for (int i = 1; i <= n; i++){
    third = first + second;
    printf("%d ", first);
    first = second;
    second = third;
  }
  return 0;
}
```

OUTPUT

```
0 1 1 2 3 5 8 13 21 34
```

INPUT

10

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QUESTIONS 18 / 18 completed

Factorial

FACTORIAL

Write a program that takes a number as input and prints its factorial using a for loop.

PROGRAM EDITOR C Run Save

```c
#include <stdio.h>
int main()
{
    int n;
    long long factorial = 1;
    scanf("%d", &n);
    for (int i = 1; i <= n; i++) {
        factorial *= i;
    }
    printf("Factorial of %d = %lld\n", n, factorial);
    return 0;
}
```

OUTPUT

```
Factorial of 5 = 120
```

INPUT

5

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QUESTIONS 18 / 18 completed

Pyramid

PYRAMID

Write a program that prints a pyramid pattern of stars using nested loops.

PROGRAM EDITOR

C Run Save

```c
#include <stdio.h>;
int main()
{
  int i, j, n;
  scanf("%d", &n);
  for (i = 1; i <= n; i++){
    for (j = 1; j <= n - i; j++)
      printf(" ");
  for (j = 1; j <= 2 * i - 1; j++){
    printf("*");
  }
  printf("\n");
}
  return 0;
}
```

OUTPUT

```
    *
   ***
  *****
 *******
*********
```

INPUT

5

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QUESTIONS 18 / 18 completed

Diamond

DIAMOND

Write a program that prints a diamond pattern of stars using nested loops.

PROGRAM EDITOR

C Run Save

```c
#include <stdio.h>
int main()
{
  int i, j, n;
  scanf("%d", &n);
  for (i = 1; i <= n; i++){
    for (j = 1; j <= n - i; j++){
      printf(" ");
    }
    for (j = 1; j <= 2 * i - 1; j++){
      printf("*");
    }
    printf("\n");
  }
  for (i = n - 1; i >= 1; i--){
    for (j = 1; j <= n - i; j++){
      printf(" ");
    }
    for (j = 1; j <= 2 * i - 1; j++){
      printf("*");
    }
```

OUTPUT

```
    *
   ***
  *****
 *******
*********
 *******
  *****
   ***
    *
```

INPUT

5

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QUESTIONS 18 / 18 completed

R- triangle

R- TRIANGLE

Write a program that prints a right-angled triangle pattern of stars using nested loops.

PROGRAM EDITOR

C

Run Save

```c
#include <stdio.h>
int main()
{
  int i, j, n;
  scanf("%d", &n);
  for (i = 1; i <= n; i++){
    for (j = 1; j <= i; j++){
      printf("*");
    }
    printf("\n");
  }
  return 0;
}
```

OUTPUT

```
*
**
***
****
*****
```

INPUT

5

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QUESTIONS 18 / 18 completed

Hollow Square

HOLLOW SQUARE

Write a program that prints a hollow square pattern of stars using nested loops.

PROGRAM EDITOR

C

Run Save

```c
#include <stdio.h>
int main()
{
   int i, j, n;
   scanf("%d", &n);
    for (i = 1; i <= n; i++) {
        for (j = 1; j <= n; j++) {
            if (i == 1 || i == n || j == 1 || j == n)
                printf("*");
            else
                printf(" ");
        }
        printf("\n");
    }
    return 0;
}
```

OUTPUT

```
****
*  *
*  *
****
```

INPUT

4

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QUESTIONS 18 / 18 completed

Hollow Diamond

HOLLOW DIAMOND

Write a program that prints a hollow diamond pattern of stars using nested loops.

PROGRAM EDITOR

C | Run | Save

```c
#include <stdio.h>
int main()
{
    int i, j, n;
    scanf("%d", &n);
    for (i = 1; i <= n; i++) {
        for (j = 1; j <= n - i; j++) {
            printf(" ");
        }
        for (j = 1; j <= 2 * i - 1; j++) {
            if (j == 1 || j == 2 * i - 1 || i == n)
                printf("*");
            else
                printf(" ");
        }
        printf("\n");
    }
    for (i = n - 1; i >= 1; i--) {
        for (j = 1; j <= n - i; j++) {
            printf(" ");
        }
```

INPUT

5

OUTPUT

```
    *
   * *
  *   *
 *     *
*********
 *     *
  *   *
   * *
    *
```

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QUESTIONS 18 / 18 completed

Pascal's triangle

PASCAL'S TRIANGLE

Write a program that prints a Pascal's triangle pattern using nested loops.

PROGRAM EDITOR

C | Run | Save

```c
#include <stdio.h>
int main() {
    int rows, coef = 1, space, i, j;
    scanf("%d", &rows);
    for (i = 0; i < rows; i++) {
        for (space = 1; space <= rows - i; space++) {
            printf("  ");
        }
        for (j = 0; j <= i; j++) {
            if (j == 0 || i == 0)
                coef = 1;
            else
                coef = coef * (i - j + 1) / j;
            printf("%4d", coef);
        }
        printf("\n");
    }
    return 0;
}
```

INPUT

5

OUTPUT

```
               1
           1       1
       1       2       1
   1       3       3       1
1      4       6       4       1
```

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QUESTIONS 18 / 18 completed

zigzag

ZIGZAG

Write a program that prints a zigzag pattern of stars using nested loops.

PROGRAM EDITOR

C

Run Save

```c
#include <stdio.h>
int main()
{
    int i, j, n;
    printf("Enter the number of colums: \n");
    scanf("%d", &n);
    for(i = 0; i < 3; i++)
    {
        for(j = 0; j < n; j++)
        {
            if((i == 0 && j % 4 == 2) ||
               (i == 1 && (j % 4 == 1 || j % 4 == 3)) ||
               (i == 2 && j % 4 == 0))
            {
                printf("*");
            }
            else
            {
                printf(" ");
            }
        }
```

INPUT

13

OUTPUT

```
Enter the number of colums:
  *   *   *
 * * * * * *
*   *   *   *
```

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QUESTIONS 18 / 18 completed

Spiral pattern

SPIRAL PATTERN

Write a program that prints a spiral pattern of numbers using nested loops.

PROGRAM EDITOR

C

Run Save

```c
#include <stdio.h>
int main()
{
  int n;
  scanf("%d", &n);
  int top = 0, bottom = n - 1, left = 0, right = n - 1;
  int i, j, num = 1;
  int a[n][n];
  while (top <= bottom && left <= right){
    for (j = left; j <= right; j++)
      a[top][j] = num++;
    top++;
    for (i = top; i <= bottom; i++)
      a[i][right] = num++;
    right--;
    if (top <= bottom){
      for (j = right; j >= left; j--){
        a[bottom][j] = num++;
      }
      bottom--;
    }
```

INPUT

5

OUTPUT

```
  1   2   3   4   5
 16  17  18  19   6
 15  24  25  20   7
 14  23  22  21   8
 13  12  11  10   9
```

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QUESTIONS 18 / 18 completed

Armstrong

ARMSTRONG

Write a program to find the given number is Armstrong or Not.

PROGRAM EDITOR

C

Run Save

```c
#include <stdio.h>
int main()
{
  int n, temp, rem, sum = 0;
  scanf("%d", &n);
  temp = n;
  while (temp != 0){
    rem = temp % 10;
    sum = sum + rem * rem * rem;
    temp = temp / 10;
  }
  if (sum == n)
    printf("Amstrong Number");
  else
    printf("Not an Amstrong Number");
  return 0;
}
```

OUTPUT

```
Amstrong Number
```

INPUT

153

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QUESTIONS 18 / 18 completed

Perfect

PERFECT

Write a program to find the given number is Perfect number or Not.

PROGRAM EDITOR

C

Run Save

```c
#include <stdio.h>
int main()
{
  int n, i, sum = 0;
  scanf("%d", &n);
  for (i = 1; i < n; i++){
    if (n % i == 0){
      sum = sum + i;
    }
  }
  if (sum == n)
    printf("Perfect Number");
  else
    printf("Not a Perfect Number");
  return 0;
}
```

OUTPUT

```
Perfect Number
```

INPUT

28